

# Antoine Lorient

DATA SCIENTIST, APPLIED MACHINE LEARNING FOR HUMAN AND COMPUTER INTERACTION

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## Research

### CNRS - IRCAM - INRIA

Paris, France

POST-DOCTORAL RESEARCHER

Jan. 2020 - Sept. 2022 (2.5 years)

- Studied the perception of motor learning in tasks involving complex and creative motions by implementing a metric learning approach on top of Dynamical Time Warping (Pytorch) - published in *PLOS One*.
- Improved upon and released an open-source project on Github and Pypi implementing the soft-DTW algorithm for CUDA devices (Pytorch, CUDA) now downloaded more than 2k times - available at *pysdtw*.
- Optimised a training schedule for a motor task using reinforcement learning and MABs based on a framework specialised in interactive machine learning (Marcelle, Javascript, Python) - submitted to IHM'22.
- Developed a prototype for movement exploration and modelling based on an algorithm performing live incremental clustering of time series (Javascript, MaxMSP, Python, C++) - demoed at NIME'22.

### University of Glasgow

Glasgow, United Kingdom

RESEARCH ASSISTANT AND PHD CANDIDATE

Mar. 2015 - Sep. 2019 (4.5 years)

- Researched *computational methods for the study of novel gestural interactions*. Deployed discriminative models to create virtual surfaces from RGB-D sensors - published at *ISS'17*; optimised gameplay experiences for custom game controllers using user probabilistic modelling - available on *arXiv*; explored and characterised user physical capabilities with audio-based reinforcement learning.
- Contributed to the European project MoreGrasp: led redaction of deliverable documents, implemented visual feedback mechanisms on smartwatches (Android), prepared live demos and posters for review meetings, established contacts with health practitioners at Glasgow's hospital leading to workshops and a funding proposal.
- Analysed visualisation techniques (*Jupyter notebooks*) for low-dimensional embeddings, i.e. t-SNE.
- Tutored the courses Artificial Intelligence and Data Fundamentals for Undergraduate and Master students.
- Developed and released open source software for Intel Realsense camera. This package was recommended by Intel as a primary choice for community maintained Python bindings (until librealsense 2.0 was released) and has accrued over one hundred stars on github and 45k downloads to this day - available at *pyrealsense*.

### Stockholm University, Mobile Life

Stockholm, Sweden

RESEARCH ASSISTANT

Feb. 2014 - Oct. 2014 (9 months)

- Explored innovative designs for always-on in the background speech interactions - published at *CHI'15*. Ran an experiment capturing day-long recordings of participants' everyday life followed by qualitative analysis and design workshop with trained Interaction designers.

## Industry

### Now & Net

Paris, France

FOUNDER

2013 - 2015 (2 years)

- Co-founded a startup providing a collective musical experience organised around DJ-ing.
- Developed a proof-of-concept recommender system (Neo4j) for creating attractive playlists.
- Wrote R&D plans leading to the successful application to the status of JEI.

### Ericsson AB

Stockholm, Beijing, Seattle

MULTIPLE POSITIONS

2008 - 2013 (6 years)

- SCRUM master and Software Developer [2013/2014]: Acted as the "guardian of the process" to facilitate daily operations of a cross functional team (8 people). Animated planning meetings, daily follow-up meetings and retrospective meetings.
- System engineer AT&T laboratory [summer 2012]: Integrated features into the customer network with laboratory and in-the-field experiments.
- Research engineer at Ericsson Research [2012]: Prepared the 4G+ prototype for the Mobile World Congress (MWC).
- Software Developer 4G/LTE [2008/2011]: Developed algorithms (embedded C) from standard specifications (3GPP). Extended and supported an Erlang test framework used by 50 engineers. Provided training sessions in Stockholm and Beijing.

## Education

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### University of Glasgow

*Glasgow, United Kingdom*

PHD IN COMPUTING SCIENCE

*grad. 2019*

Thesis title: A Computational Approach to Gestural Interactions of the Upper Limb on Planar Surfaces.

### Royal Institute of Technology (KTH)

*Stockholm, Sweden*

MSC IN APPLIED PHYSICS

*grad. 2009*

### Telecom Physique Strasbourg

*Illkirch-Graffenstaden, France*

MSC IN ENGINEERING AND PHYSICS

*grad. 2008*

## Skills

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| <b>Data Modelling</b>   | Python, Pytorch, Keras, Cuda, MIFlow, Pyro, Seaborn, Jupyter                     |
| <b>Applied Research</b> | Scientific Writing, Quantitative Analysis, Experiment Design, Ethics             |
| <b>Soft skills</b>      | Lean, Agile, Scrum Master, Teaching, Tutoring, English, French (native), Swedish |